Smart Restaurant (Cinnamon Bistro) — Microservice Testing Pipeline

SE3112 Advanced Software Engineering · take-home assessment · system under test: SE3022 microservices

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SE3022 ARCHITECTURE (SYSTEM UNDER TEST)

React 19 + Vite SPA (Nginx)

▼ /api ▼ /reservation-api · Bearer JWT

Identity service
ASP.NET Core 10 · :5001
register · login · BCrypt · JWT · profiles

Reservation service
ASP.NET Core 10 · :5000
tables · availability · bookings · reports

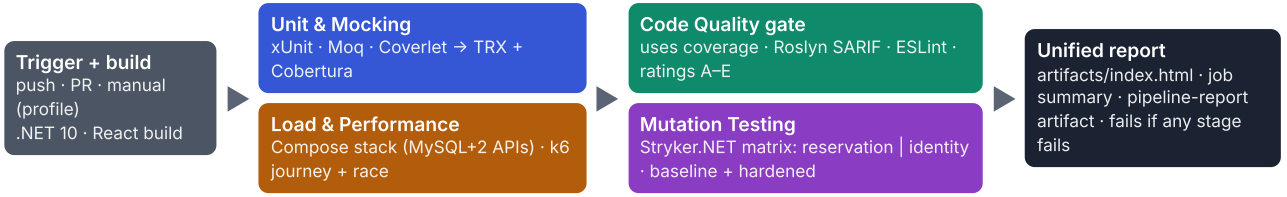
▼ ADO.NET ▼ ADO.NET · row locks · outbox

MySQL identity_db

MySQL reservation_db → Kafka (outbox)

Shared JWT contract (issuer, audience, NameIdentifier claim). Double booking is prevented with `SELECT … FOR UPDATE` and a half-open overlap check; lifecycle events go through a transactional outbox.

UNIFIED TEST AUTOMATION — GITHUB ACTIONS `TEST-PIPELINE.YML` · LOCAL TWIN `SCRIPTS/RUN-PIPELINE.SH`



ROLE / OWNER	IMPLEMENTATION	KEY RESULT
Unit Testing & Mocking H.R.M.A.A. Bandara	184 tests: `[Theory]` with InlineData / generated MemberData (full 4×4 state machine) / ClassData; strict mocks, stubs with `out` params, spies via Callback, fakes (clock, in-memory repo)	414 tests green · 68.6% line / 68.1% branch coverage
Mutation Testing H. L. P. S. Perera	Stryker.NET (*Complete* level, per-test coverage, break < 60%) on domain rules + identity services; survivors analysed → 23 targeted hardening tests	Reservation 81.5% → **97.5%** · Identity 82.2% → **100%** · 3 equivalent mutants (dead branch)
Load & Performance D.M.N. Pesanjith	k6 closed-model journey (browse → search → book → view → cancel → history) + open-model login; smoke / load / stress / spike / soak; per-endpoint SLO thresholds; 50-VU double-booking race	Load: all SLOs ✓, 0% errors, booking p95 9 ms · Stress: BCrypt login knee ≈ 25/s · Race: 1 winner / 49 × 409 per round
Code Quality & Metrics Wijesinghe K.	Opt-in Roslyn analyzers (complexity CA1502, coupling CA1506, SQL CA2100…) → SARIF; ESLint flat config; SQALE debt model; hotspot review; ratchet gate	Gate ✓ · Maintainability A (0.62%) · Security A (4 hotspots SAFE) · Reliability C · 122 smells ≈ 3 d debt

TOOL CHOICES & WHY

- **xUnit + Moq**: already the SE3022 stack; Moq supports strict mocks, ref/out stubs and callbacks.
- **Stryker.NET**: the de-facto .NET mutation tool; HTML report shows each mutant in the source.
- **k6**: scripts are code (versioned, reviewable), thresholds act as SLO gates, and it runs headless in CI.
- **Roslyn analyzers + custom gate**: SonarQube-style metrics without a server or secret token.

AUTOMATION DESIGN DECISIONS

- One script, two runners: CI calls `run-pipeline.sh`, so local demo = CI behaviour.
- Stages run in parallel where independent; quality reuses the unit-test coverage artifact.
- Disposable load stack on tmpfs MySQL; the Kafka publisher is disabled (outbox still written).
- Gate philosophy: hard limits for security/coverage, a *ratchet* for legacy smells (clean as you code).

RUN IT

- `./scripts/run-pipeline.sh all` → `artifacts/index.html`
- `… unit` · `… quality` · `… mutation baseline|hardened`
- `… load smoke|load|stress|spike|soak` · `… race`
- GitHub → Actions → *SE3112 Test Pipeline* → Run workflow (choose profile)

Repository: PraveenSanjaya/Smart-Restaurant-Management-System

Details: README.md § SE3112 and tests/*/README.md